operation, under the assumption that keys' lengths are bounded by a constant.

4.10 Estimating Bloom Filter Precision[19]

Before we start, we need to fix some notation and make a few more assumptions:

- `m` is the number of bits in our array.
- `k` is the number of hash functions we use to map a key to `k` different positions in the array.
- Each of the `k` hash functions we use is independent from the others.
- The pool of hash functions from which we extract our `k` functions is a universal hashing set.

If these assumptions hold, then it can be proven that a good approximation for the false positive probability after `n` insertions is given by

$$p\left(n, m, k\right) = \left(1 - e^{-\frac{k \cdot n}{m}}\right)^{k}$$

where `e` is Euler's number, the base of natural logarithms.

Now we have a formula to estimate the probability of a false positive! That's nice *per se*, but what's even better is that it allows us to tune `m` and `k`, our Bloom filters parameters: this